Diesel Engines
V 4000 R 10
V 4000 R 20

Maintenance Schedule
M050478/01E

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept.
Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.
The Maintenance Schedule is based on the power output defined in the individual orders, under consideration of the operating parameters established in the order's specifications.
The time intervals according to which the maintenance tasks and the relevant checks and tasks involved are to be completed are based on operational experience and can only serve as guidelines. Specific operational conditions may require modification of the Maintenance Schedule.

Note:

The engine manufacturer cannot be held responsible for the performance of the following maintenance tasks.
The operator is responsible for carrying out the following maintenance tasks:
Fuel filter
The maintenance interval depends on the degree of fuel contamination. The fuel filter must be replaced after 2 years at the latest.
Battery
Battery maintenance depends on the degree of use and ambient temperature. The battery has to be checked according to the manufacturer's instructions, as it is not always a part of the MTU scope of supply.
Fluids and Lubricants:
Only those fluids and lubricants are to be used that are approve of in the MTU Fluids and Lubricants Specification A001061. A special note must be made that, after maintenance or repair work on the cooling system, the operator must check the concentration of the corrosion inhibitor or the corrosion-inhibitor antifreeze.

Major Overhaul

Requires complete engine disassembly to facilitate repair/overhaul of all relevant components.

Out-of-Service Periods

If the engine is to remain out of service for several weeks, spray engine oil into the air-intake ducting. Actuate the engine shutdown system and bar the engine with the starting equipment when performing this task. Carry out maintenance tasks in accordance with the time limit schedule (see the Maintenance Frequency Chart). In addition, the engine must be barred manually every month. If the engine is to remain out of service for more than 1 month, carry out engine preservation in accordance MTU Fluids and Lubricants Specification, Publication No. A001061.

Application group

2A: Rail Traction

Maintenance Frequency Chart

Maintenance Level	W1:	Operational checks, daily	X
	W2:	Operating hours	500
		Limit, months	6
	W3:	Operating hours	1 000
		Limit, year	1
	W4:	Operating hours	5 000
		Limit, year(s)	2
	W5:	Operating hours	10 000
		Limit, year(s)	6
	W6:	Operating hours	30 000
		Limit, year(s)	18

Maintenance Task Schedule

Code No.	Maintenance Level W1 – Operational checks	
G000.000.13	Engine operation	Check speed
G000.000.15	Engine operation	Check pressures (if gauges are provided)
G000.000.17	Engine operation	Check temperatures (if gauges are provided)
G000.000.19	Engine operation	Check running noises
G000.000.21	Engine operation	Check engine and external pipework for leaks
G101.011.05	Turbocharger connections	Check exhaust connections for leaks Check oil supply/return pipework for leaks
G121.052.01	Air filter	Check service indicator
G124.051.01	Intake air line	Check drain line for water discharge and obstructions
G140.000.05	Exhaust system	Check exhaust gas color
G140.000.11	Exhaust system	Drain condensate (if drain valve provided)
G180.000.01	Engine oil	Check level
G202.000.03	Engine coolant	Check level
G510.070.01	Monitoring system	Carry out lamp test

	Maintenance Levels		W	2	3	4	5	6
G120.000.03	Intake air line	Check air ducting for leaks and damage						
G123.051.01	Air-flow control flap	Check ease-of-movement						
G144.051.01	Exhaust flow control flap	Check ease-of-movement						
G180.000.04	Engine oil	Take sample and analyze						
G180.000.05	Engine oil	Change – With oil category 1, with centrifugal oil filter Every 500 operating hours – With oil category 2, with centrifugal oil filter Every 1000 operating hours – With oil category 3, with centrifugal oil filter Every 1500 operating hours After 2 years at the latest						
G183.052.03	Engine oil filter	Replace – Accomplish when changing the oil						
G183.101.01	Centrifugal oil filters	Clean, check thickness of oil sludge layer Replace paper sleeve						
G055.050.01	Valve gear	Check valve clearances						
G083.051.03	Fuel duplex filter	Replace filter element						
G083.051.03	Fuel duplex filter	Change throw-away filter						
G083.101.15	Fuel pre-filter	Clean						
G121.051.03	Air filter	Clean, empty dust collection box						
G140.000.11	Exhaust system	Check, check drain for obstructions						
G180.000.05	Engine oil	Change – with oil category 2						
G183.052.03	Engine oil filter	Replace – Accomplish when changing the oil						
G202.000.05	Engine coolant	Take sample and analyze (Change engine coolant, see MTU Fluids and Lubricants Specification)						
G202.051.01	Engine coolant pump	Check relief bore for obstructions						
G203.000.07	Charge air coolant	Take sample and analyze, change if necessary (see MTU Fluids and Lubricants Specification A001061)						
G203.051.01	Charge air coolant pump	Check relief bore for obstructions						
G205.051.01	Engine-coolant cooler (HT circuit)	Check cooler elements for external contamination						
G205.051.03	Engine-coolant cooler (LT circuit)	Check cooler elements for external contamination						
G365.000.03	Exhaust system	Check, check drain for obstructions						
G018.101.02	Crankcase ventilation	Replace filter elements						
G073.051.07	HP fuel delivery pump	Check relief bore for obstructions						
G075.051.05	Fuel injector(s)	Replace						
G121.051.01	Air filter	Replace filter elements						
G213.051.01	Charging generator	Check condition of coupling						
G223.051.01	Hydraulic pump drive (equipment rack)	Check condition of coupling						
G223.052.01	Hydraulic pump drive (crankshaft)	Check condition of coupling						
G231.051.01	Engine mounts	Check tightness of securing screws and nuts						
G231.051.03	Engine mounts	Check condition of resilient mounts						
G231.051.07	Engine mounts	Check buffer clearances						
G252.051.01	Coupling (Küsel)	Check radial clearance						
G500.000.09	Engine wiring:	Check						
G508.000.01	ECU	Check (visual inspection)						
G508.000.02	ECU	Clean						
G510.000.01	Monitoring system	Check function of monitoring equipment						
	Prior to commencing W5 maintenance, run the engine and record all measured values, then drain engine coolant and flush cooling system!							
	Valve gear	Remove and check rocker arms and valve bridges Check swing followers, rollers, ball sockets and swing follower bearings for wear						
	Cylinder heads	Replace						
	Running gear	Check piston crowns (visual inspection)						
	Crankcase	Check cylinder liner wear pattern						
	Vibration damper	Replace						
	Turbocharger	Replace						

	Air ducting	Remove, clean and replace seals		
	TC control	Check actuating cylinder for ease-of-movement and linkage for wear		
	Air-flow control flap	Replace air-flow control flap stops, bearing and gasket		
	Exhaust flow control flap	Remove, clean and check flap, drive shaft and flap bearings		
	Intercooler	Replace		
	Exhaust silencer	Clean, replace seals		
	HP fuel delivery pump	Replace		
	HP fuel delivery-pump sensor	Replace		
	Engine coolant pump	Replace		
	Charge air coolant pump	Replace		
	Engine-coolant cooler (HT circuit)	Clean and leak test		
	Engine-coolant cooler (LT circuit)	Clean and leak test		
	Coolant thermostat (HT circuit)	Check, replace if necessary		
	Coolant thermostat (LT circuit)	Check, replace if necessary		
	Engine coolant preheating	Overhaul		
	Engine oil heat exchanger	Remove, clean, leak test and replace seals		
	Crankcase ventilation	Replace filter element		
	Centrifugal oil filter	Check, replace bearings if necessary		
	Coupling (Küsel)	Disassemble, check parts, replace friction ring and friction disk		
	Major Overhaul	Total disassembly of the engine; inspection / repair of all components		

